

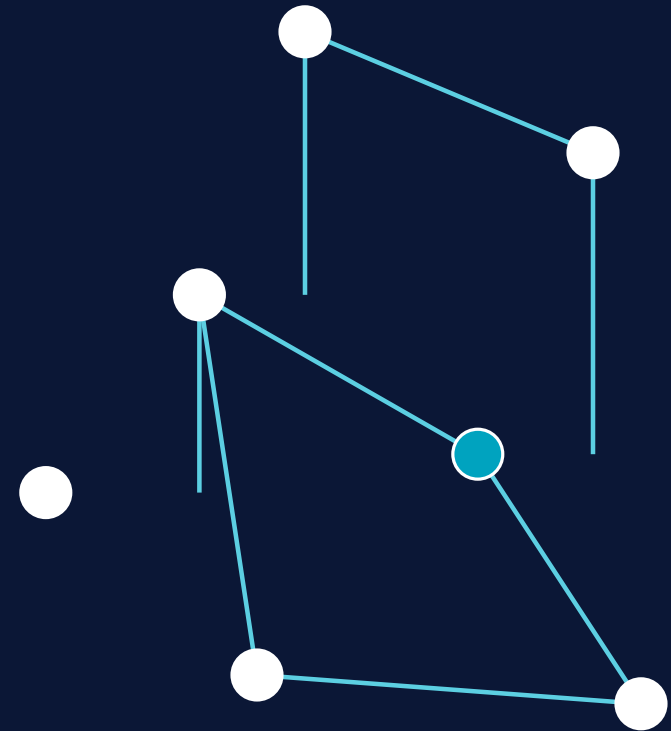
NetSage AI

Applied AI + Network Troubleshooting

Final Project Evaluation Presentation

Aditya Raj

Roll No. 25SCS1003000992



School of Computer Science & Engineering | IILM University

From coursework to an applied AI project

The internship was structured in two phases: foundation learning followed by hands-on project work.

PHASE 1

Foundation learning

- Data Analytics Essentials
- Introduction to Modern AI
- Apply AI: Analyze Customer Reviews

PHASE 2

NetSage AI project

- AI-assisted network troubleshooting
- Python rule-based validation
- Human review + dashboard analysis

The report explicitly connects the three courses to the practical NetSage AI project.

Network symptoms rarely point to one obvious root cause

Junior engineers may know individual Cisco commands but still struggle to connect a symptom to the actual fault.

Example symptom

“PC receives an IP but cannot reach a server.”

AMBIGUITY

VLAN • Gateway • DHCP • DNS • Routing •
ACL • NAT

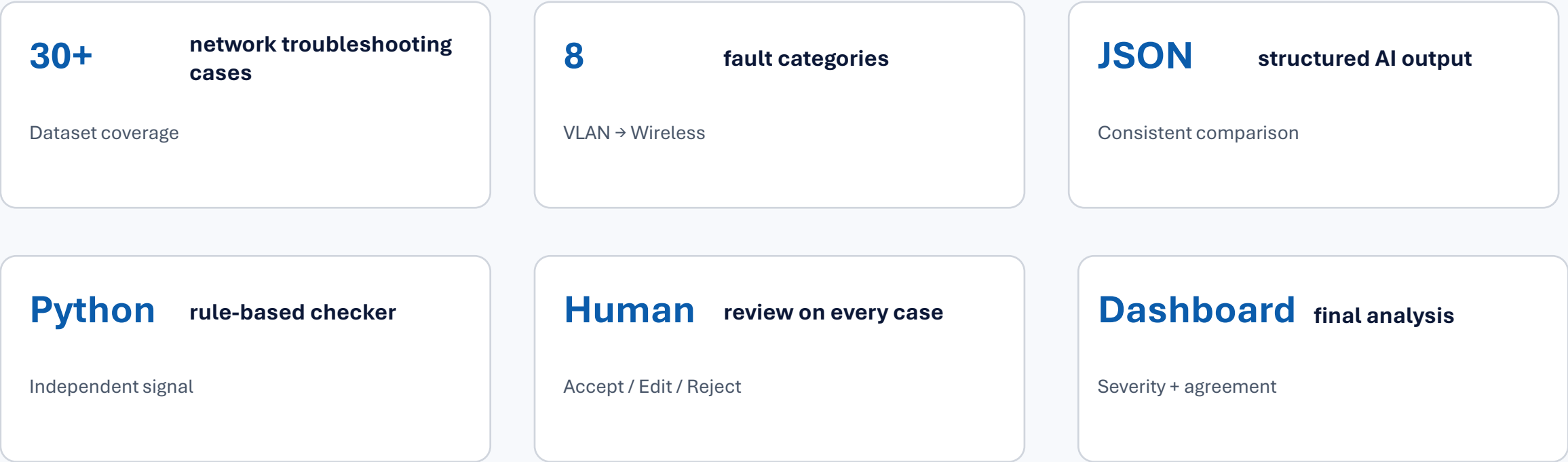
What the project needed to do

- Use the symptom, topology note and show-command output as evidence.
- Suggest the likely root cause, OSI layer, next diagnostic command and fix.
- Keep a human reviewer in the loop before accepting any diagnosis.

The system is a troubleshooting support tool—not an autonomous network configuration system.

Build a structured, evidence-based troubleshooting workflow

The project goals were defined around data coverage, AI consistency, independent validation, review, and reporting.



The report specifies at least 30 cases; the completed project contains exactly 30 cases.

What NetSage AI covers—and what it does not

The project targets Cisco-style Packet Tracer lab troubleshooting, using provided evidence to support diagnosis.

Fault categories

VLAN

Gateway

DHCP

DNS

Routing

ACL

NAT

Wireless

Boundary

Decision support —not autonomous change

The AI proposes.
The rule checker cross-checks.
The human reviewer decides.

A sequential pipeline with multiple checks

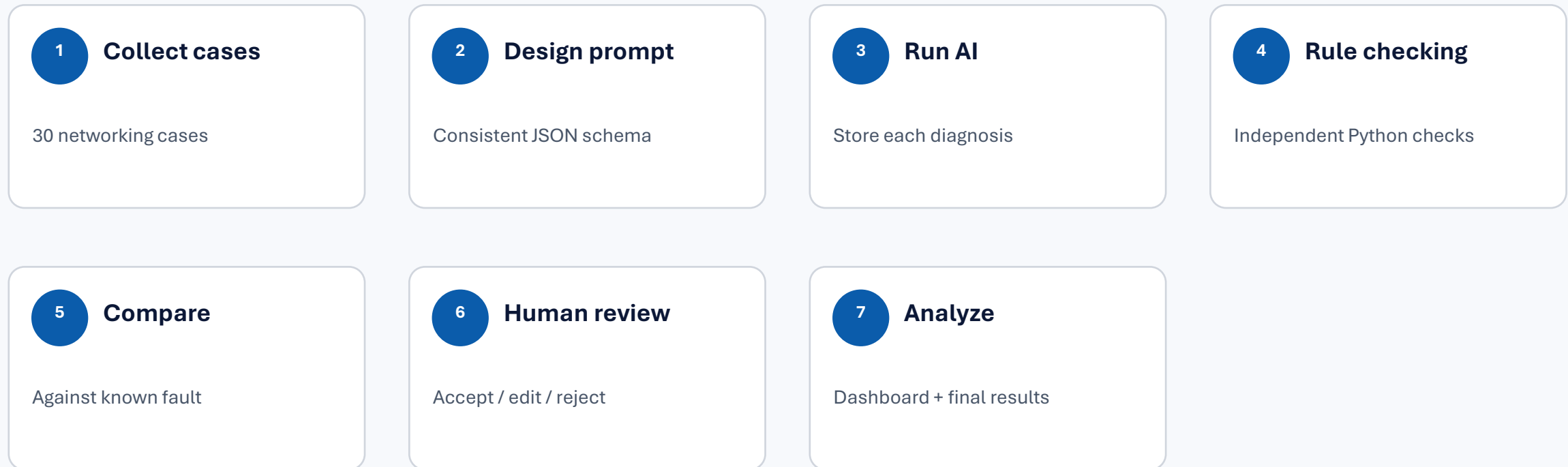
Each troubleshooting case passes through AI diagnosis, deterministic validation, comparison and human review before final reporting.



Architecture summarized from the report's "Case Data → AI Diagnosis → Rule Checker → Comparison → Human Review → Dashboard / Final Result" flow.

Seven steps from case collection to final analysis

The process was designed to make AI outputs comparable, reviewable and auditable.



Tools were combined around one practical workflow

The report lists AI, prompt engineering, Python, Packet Tracer concepts, spreadsheets, CSV and JSON as the main implementation pieces.

AI

Diagnosis from network evidence

Prompt Engineering

Consistent structured response

Python

Independent rule-based checks

Packet Tracer concepts

Cisco-style lab evidence

Excel / spreadsheets

Analysis + dashboard

CSV

Case + review data

JSON

Standardized AI output

Every response is structured for review and comparison

A fixed JSON-style schema was used so outputs could be evaluated consistently across cases.

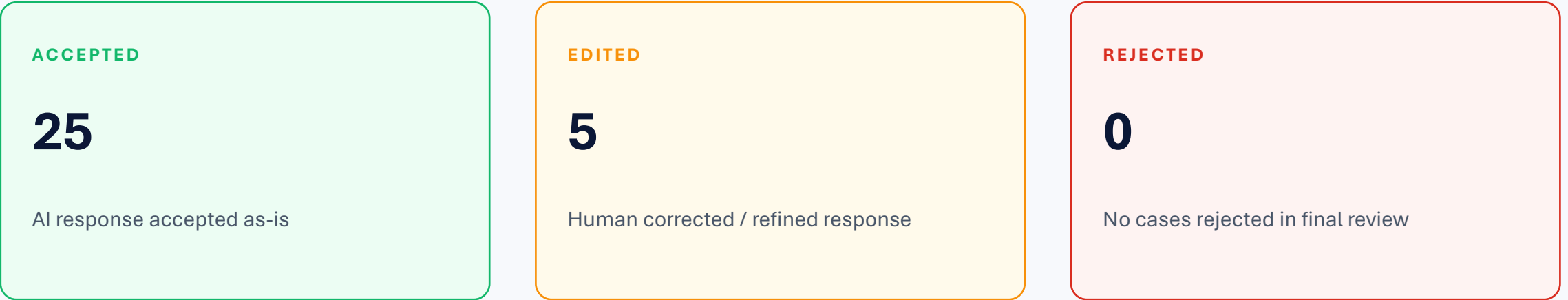
```
{  "root_cause": "...",  
  "osi_layer": "...",  
  "confidence": "...",  
  "evidence": [...],  
  "next_command": "...",  
  "fix_steps": [...]  
}
```

Why this structure matters

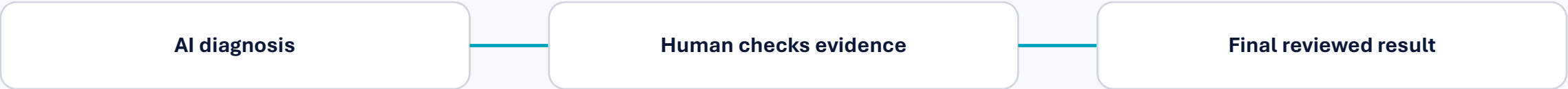
- Separates diagnosis from evidence.
- Makes confidence explicit instead of implicit.
- Creates a consistent object for comparison and review.
- Supports downstream spreadsheet/dashboard analysis.

Human review is part of the system—not an afterthought

Every AI diagnosis is reviewed before it is considered trustworthy. Corrections are recorded in the human-review log.

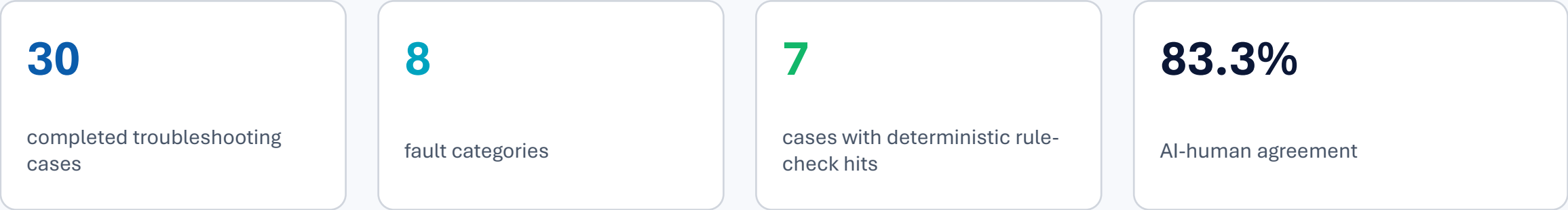


Review decision flow



30 cases across 8 troubleshooting categories

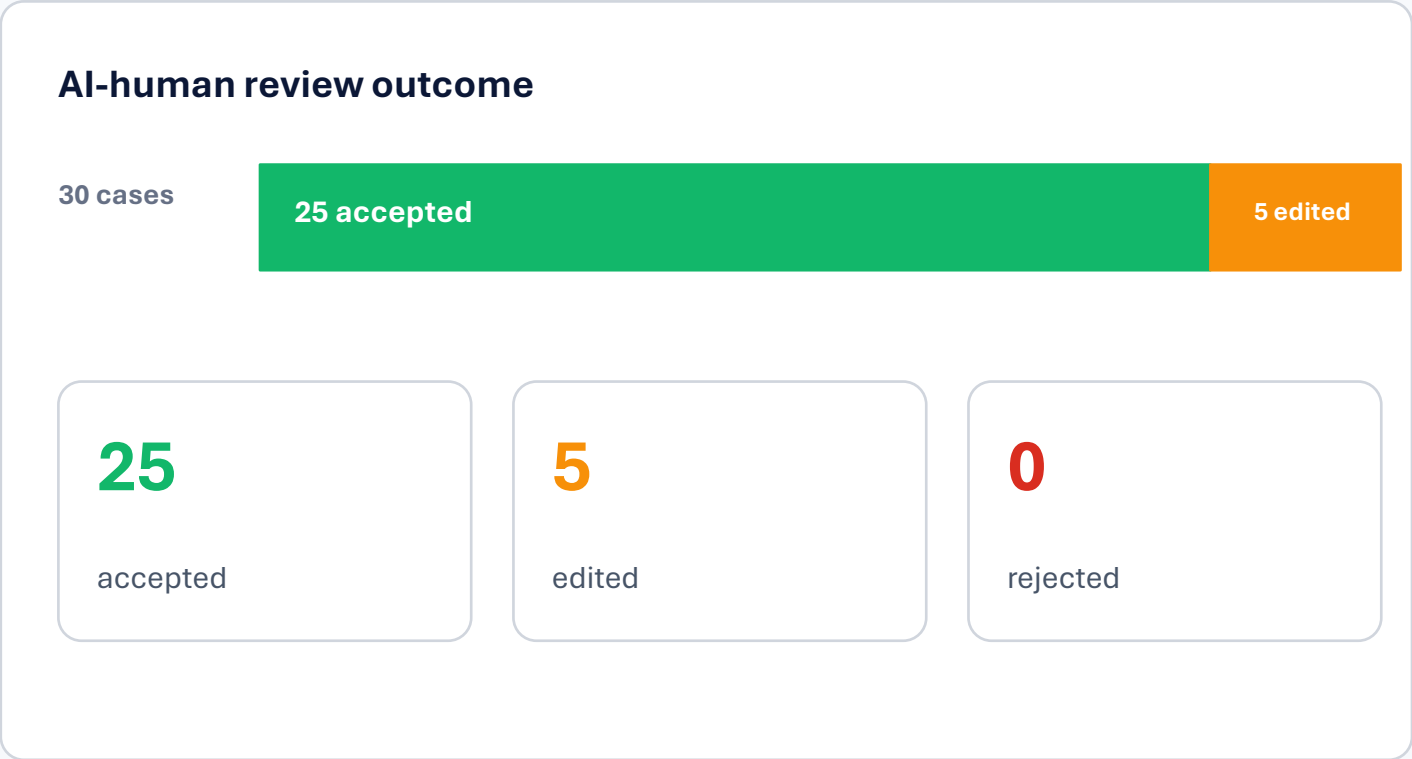
Each case includes the symptom, topology note, show-command output, expected fault, OSI layer, concept and severity.



The 83.3% figure is calculated in the report from 25 accepted responses out of 30 cases.

The strongest evidence is the reviewed outcome—not raw AI output

The final review produced 25 accepted and 5 edited responses, with no rejected cases.



83.3%

AI-human agreement

7 / 30

cases with a deterministic issue detected by the rule checker

What the internship demonstrated

The project connected AI concepts, data handling, Python validation and network troubleshooting into one applied workflow.

Project outcomes

- Structured troubleshooting support instead of isolated command lookup.
- Evidence-based responses grounded in the supplied case data.
- Independent deterministic checks to complement AI output.
- Documented human corrections for responsible AI use.

Skills gained

Artificial Intelligence

Prompt design

Python

Networking troubleshooting

Data organization

Responsible AI

Overall result

A human-supervised decision-support workflow for lab networking.

NetSage AI turns troubleshooting into a reviewable workflow

The project's value is not “AI replaces the engineer”; it is that AI, deterministic checks and human judgment work together.

AI

Generate structured diagnosis

RULES

Cross-check mechanical faults

HUMAN

Approve or correct

Final takeaway

Evidence first. AI assists. Humans remain accountable.

THANK YOU

Questions?

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Aditya Raj

B.Tech CSE • IILM University • Cisco VIP Internship Program

Final Project Evaluation